Rapid Parameter Inference for Spatiotemporal Stochastic Biological Models using Neural Posterior Estimation

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Abstract

Cell migration is a fundamental biological process underlying wound healing, tissue development, and cancer metastasis, yet calibrating mathematical models of migration to experimental data remains a major challenge. Scratch and barrier assays are widely used to study collective cell spreading, and agent-based random walk models provide a natural stochastic description of these experiments. However, parameter inference for such models is hampered by intractable likelihoods, forcing researchers to rely on Approximate Bayesian Computation, which introduces biases and tuning difficulties, or surrogate models that require potentially erroneous noise model specifications. Here, we overcome these limitations using Neural Posterior Estimation (NPE), a simulation-based inference framework that learns the full posterior distribution directly from stochastic simulations. Once trained, NPE provides amortized inference, where the upfront cost of training is offset by near-instant posterior estimates for any new observation. We deploy this framework on a random walk model of a barrier assay experiment describing in vitro cell migration and demonstrate our approach in two settings: first, using one-dimensional summary statistics (column counts), and second, by augmenting NPE with a convolutional neural network (CNN) that enables inference directly from raw two-dimensional spatial data. In both cases NPE proves highly effective, delivering posterior estimates at a substantially reduced computational cost per inference. When applied to one-dimensional summary statistics, the pipeline matches or exceeds the precision of classical methods; for two-dimensional spatial data, NPE augmented with a CNN performs inference directly on raw images with a precision comparable to that from curated summary statistics. This capability is critical as it moves beyond reliance on lower-dimensional data and opens new avenues for complex models where informative summary statistics are not obvious or available. We provide an open-source implementation of our pipeline to facilitate its adoption and extension to more complex, spatially-structured biological models.

1 Introduction

Cell migration is a fundamental biological process that drives wound healing, embryonic development, immune responses, and cancer metastasis [**? ?**]. A quantitative understanding of how cell populations spread, proliferate, and interact requires calibrating mathematical models to experimental data, and scratch and barrier assay experiments have become key tools for studying collective cell behaviour in vitro [22, 23, 46]. In these experiments, cells are confined to a region on a substrate; after barrier removal, the population spreads outward, and the resulting spatiotemporal dynamics encode information about the underlying motility and proliferation mechanisms.

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Agent-based random walk models on regular lattices provide a natural stochastic description of these experiments, capturing the inherent biological variability and heterogeneity that deterministic continuum models cannot adequately represent [8, 45, 39]. However, calibrating these discrete stochastic models to data poses fundamental inferential challenges. The likelihood function is typically intractable, precluding standard statistical methods such as Maximum Likelihood Estimation (MLE) and Markov chain Monte Carlo (MCMC). Classical likelihood-free alternatives like Approximate Bayesian Computation (ABC) are computationally expensive and sensitive to tuning choices [49, 47]. A common workaround is to replace the stochastic simulator with a deterministic partial differential equation (PDE) surrogate, but this introduces systematic bias and requires specifying an explicit noise model whose misspecification can lead to poorly calibrated uncertainties [24, 6, 44]. These practical difficulties have motivated a growing interest in simulation-based inference methods that can learn directly from stochastic simulators [7, 43, 45, 42, 53, 29].

In this paper, we address these challenges using Neural Posterior Estimation (NPE), a simulation-based inference method that learns the posterior distribution $p(\theta \mid x)$ directly from parameter-data pairs generated by the stochastic simulator [35, 18, 9]. NPE is likelihood-free, learns directly from the high-fidelity simulator without surrogate approximations, implicitly captures the stochastic relationship between parameters and data without an explicit noise model, and provides amortized inference—after an initial training phase, posterior estimates for new observations are obtained almost instantaneously. A key methodological contribution is our integration of a Convolutional Neural Network (CNN) as a learned feature extractor within the NPE pipeline, enabling inference directly from raw two-dimensional spatial data and removing the need for hand-crafted summary statistics.

We demonstrate this framework on a canonical agent-based random walk model of a barrier assay experiment, using the setup of [45] (S&P2025 hereafter) as our reference. The remainder of this paper is organized as follows. Section 2 describes the random walk model, reviews classical inference approaches and their limitations, and details the NPE methodology and our CNN-based data processing pipeline. Section 3 presents inference results using both one-dimensional summary statistics and raw two-dimensional spatial data, and compares computational costs across methods. Section 4 discusses the implications of our findings for simulation-based inference in biology.

2 Methods

2.1 Random Walk Model

We study a stochastic agent-based model of a barrier assay, following the setup described in S&P2025. Agents representing biological cells are initially confined to a central region on a two-dimensional square lattice and subsequently spread outwards via an unbiased random walk. The experimental setup, illustrated in Figure 1, mimics a barrier assay where cells are placed in a confined region and allowed to spread after barrier removal. The two-dimensional lattice has height $H = 50$ (with $0 \leq y \leq 50$) and width $W = 200$ (with $-100 \leq x \leq 100$), with lattice spacing $\Delta = 1$ and discrete time step $\tau = 1$. Zero-flux boundary conditions are applied at all edges. Initial conditions are set by occupying each site in the central region $|x| \leq 25$ with a single agent with probability $U = 0.3$, as shown in Figure 1a. At each time step, the system evolves through a random sequential update method where agents move to one of four nearest-neighbour sites with probability $P = 0.7$. After 100 time steps, the population spreads symmetrically as shown in Figure 1b.

For lattice-based random walk models of this type, the mean behaviour of agents in the continuum limit can be described by a PDE of the form [8, 45, 39]:

$$\frac{\partial u}{\partial t} = -\frac{\partial J}{\partial x} + S, \tag{1}$$

where $u(x, t)$ represents mean agent density, J is the macroscopic flux, and S denotes source terms from proliferation. For the simple case considered here (unbiased motion, no proliferation, no crowding), this reduces to the linear diffusion equation:

$$\frac{\partial u}{\partial t} = D\frac{\partial^2 u}{\partial x^2}, \tag{2}$$

where $D = \lim_{\Delta,\tau\to 0} P\Delta^2/(4\tau)$ is the effective diffusion coefficient. Working with dimensionless units $\Delta = \tau = 1$, this gives $D \approx P/4 = 0.175$ for our simulation with $P = 0.7$.

While S&P2025 investigates a suite of model variations (e.g., incorporating proliferation, directional bias, and crowding effects) and alternative PDE approximations, we focus on this unbiased, non-interacting case. It provides a clear baseline against which to compare our proposed NPE method, and we refer the reader to the original study for a fuller discussion of more complex scenarios.

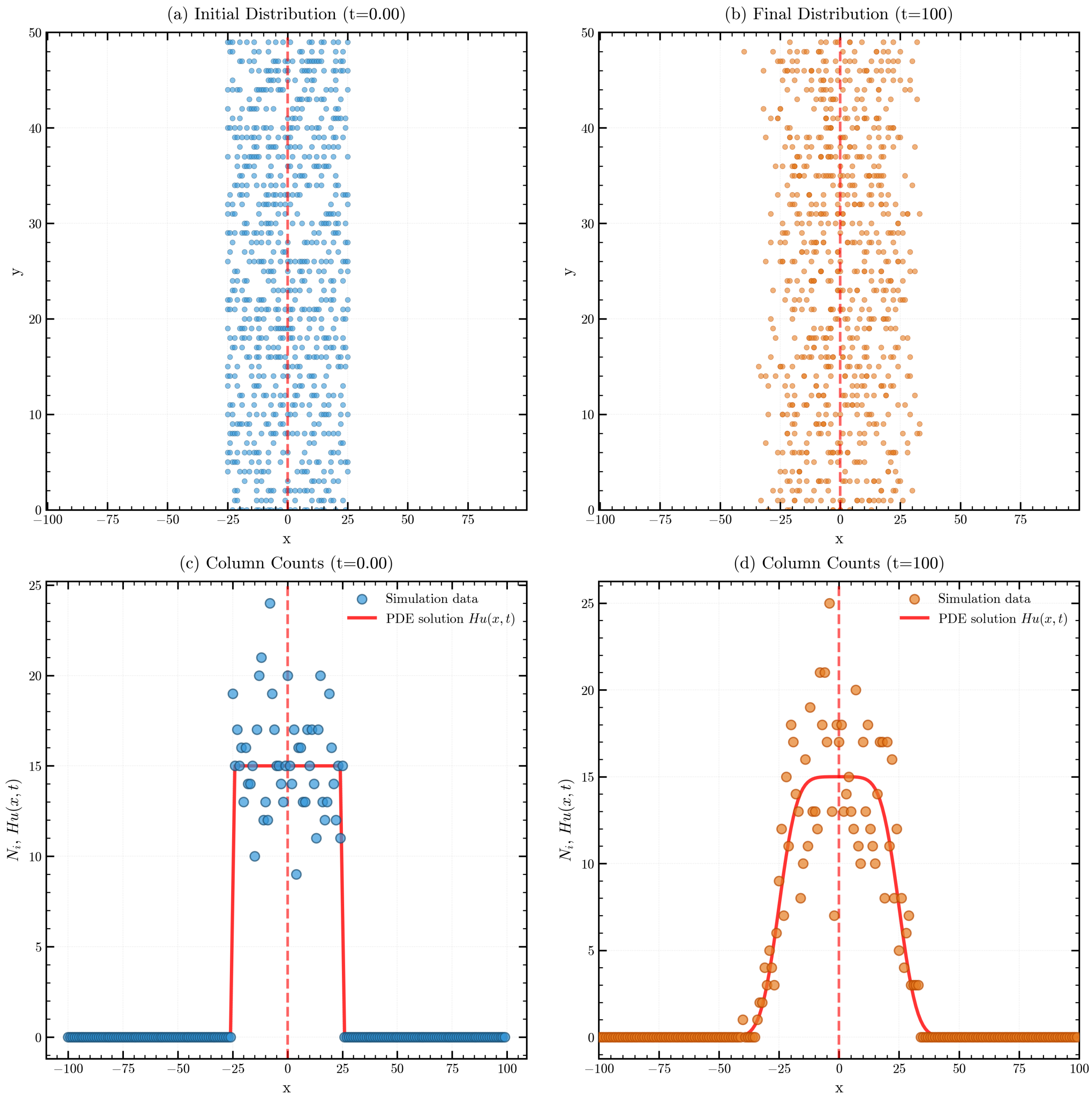


Figure 1: Schematic of the lattice-based random walk model. (a) Initial placement of agents within region $|x| \leq 25$ on a 2D lattice with $H = 50$, $\Delta = \tau = 1$, $U = 0.3$. (b) Agent distribution after 100 time steps showing symmetric outward spreading. (c)-(d) Reduction of 2D data to 1D column counts $N_i(t)$ at $t = 0$ and $t = 100$ respectively. Blue dots show initial stochastic count data; orange dots show final stochastic count data; red curves show the continuum PDE solution $Hu(x,t)$ where $u(x,t)$ solves the diffusion equation with $D = P/4$.

2.2 Classical Inference Approaches

We briefly review the established approaches to inference for stochastic agent-based models, drawing primarily on the treatment in S&P2025, which provides systematic comparisons across multiple inference paradigms for biological random-walk models. Table 1 provides a comparative overview.

Method	Cost per Inference	Requires Surrogate?	Requires Summary Stats?	Amortized?
ABC	Very High	No	Yes	No
Surrogate + MCMC	High	Yes	Yes	No
Surrogate + MLE	Medium	Yes	Yes	No
NPE (proposed)	**Low***	**No**	**Optional**	**Yes**

Table 1: Comparison of inference approaches for stochastic biological models. *The low cost per inference for NPE is achieved after an initial, computationally intensive training phase.

2.2.1 Approximate Bayesian Computation

ABC represents the most direct approach to likelihood-free inference for stochastic models [49, 47, 50, 40]. The foundational ABC rejection sampling algorithm approximates the posterior distribution by undertaking the following steps: (i) a parameter vector $\theta^{(j)}$ is drawn from the prior distribution $p(\theta)$; (ii) a dataset $Y^{(j)}$ is simulated from the model using $\theta^{(j)}$; (iii) a discrepancy between the summary statistics of the observed data, $s(Y^{\text{obs}})$, and simulated data, $s(Y^{(j)})$, is calculated using a distance metric $\rho(\cdot, \cdot)$; and (iv) the parameter $\theta^{(j)}$ is accepted if this discrepancy is less than a tolerance threshold, $\rho(s(Y^{\text{obs}}), s(Y^{(j)})) \leq \epsilon$. The collection of accepted parameters forms a sample from an approximation to the true posterior, $p(\theta \mid Y^{\text{obs}})$ [2].

Despite its conceptual simplicity, ABC faces significant methodological and computational challenges. A primary issue is the choice of the tolerance ϵ, which defines a fundamental bias-accuracy trade-off [49, 4]. Only in the idealised limit where $\epsilon \to 0$ and the summary statistics are sufficient does the ABC posterior converge to the true posterior [31, 4]. In practice, a smaller ϵ reduces bias but dramatically lowers the acceptance rate, increasing computational cost. Conversely, a larger ϵ increases the acceptance rate at the cost of yielding a more biased and diffuse posterior approximation.

This trade-off is exacerbated by the "curse of dimensionality" [3, 4]. For models with high-dimensional data or parameter spaces, the acceptance rate can be exceptionally low, demanding a vast number of simulations. For instance, S&P2025 generate 10^5 simulations from their computationally expensive random walk model to retain just the top 1%, highlighting this significant computational inefficiency. Furthermore, the method's performance depends critically on the choice of summary statistics. If the statistics are not formally sufficient, crucial information from the data is discarded, which introduces a form of bias that cannot be removed by simply lowering the tolerance ϵ [41, 31]. Selecting informative yet low-dimensional summary statistics in a principled manner remains a difficult and often ad-hoc challenge [14, 4]. More efficient variants of ABC, such as Sequential Monte Carlo ABC [48, 52], have been proposed to try to circumvent some of the above issues.

2.2.2 Surrogate Model Approaches

To mitigate the computational burden of methods like ABC, an alternative strategy replaces the stochastic model with a computationally cheaper surrogate model. A common choice is a deterministic PDE that captures the system's mean-field behaviour, yielding a tractable likelihood function [13, 1] that enables standard, efficient likelihood-based inference algorithms like MCMC. However, this computational gain comes at the cost of introducing potential model misspecification bias, as inference is performed on an approximation of the true stochastic process.

The primary limitation of this approach is that the resulting inference is conditional on the fidelity of the surrogate. Any discrepancy between the surrogate and the true stochastic process introduces a systematic bias that cannot be reduced by collecting more data [24, 6]. While the continuum limit accurately describes the mean behaviour of non-interacting random walks, the mean-field PDE neglects crucial stochastic effects in more complex scenarios. For example, in interacting models with crowding effects discussed in S&P2025 (Section 4), the mean-field approximation captures average behaviour but cannot represent the bounded nature of agent occupancy or the discrete stochastic fluctuations [5, 1] around this mean. This limitation becomes particularly important when constructing prediction intervals, where the choice of noise model (e.g. Gaussian vs. binomial, see Section 2.2.4) significantly affects the physical realism of predictions [32, 15]. These fundamental limitations of surrogate-based approaches motivate the need for inference methods that can work directly with the full stochastic simulator.

2.2.3 Inference with Surrogate Likelihoods

A tractable likelihood function from a surrogate model unlocks standard, efficient inference techniques that would be impossible with the original stochastic simulator. These techniques generally fall into two categories: direct optimization or posterior sampling.

The first category, direct optimization, seeks to find the single parameter set that maximizes the likelihood function, yielding the MLE. While computationally fast, the MLE provides only a point estimate and does not inherently quantify parameter uncertainty, although indirect (asymptotic) uncertainty quantification can be obtained through likelihood ratio tests and Wilks' theorem. The second category, posterior sampling, uses methods like MCMC to approximate the full posterior distribution, offering a complete characterisation of uncertainty. However, MCMC typically requires significant computational effort, careful hyper-parameter tuning, and rigorous convergence assessment. A third approach, the Laplace approximation, offers a compromise between the speed of optimization and the detailed uncertainty characterization of MCMC. This method approximates the posterior distribution as a multivariate normal distribution centred at the posterior mode (which equals the MLE, $\hat{\theta}$ under uniform priors). The covariance is given by the inverse of the observed Fisher Information matrix:

$$p(\theta \mid Y^{\text{obs}}) \approx \text{MVN}(\hat{\theta}, \mathcal{I}^{-1}(\hat{\theta})), \tag{3}$$

where $\mathcal{I}(\hat{\theta}) = -\nabla\nabla^T \ell(\theta \mid Y^{\text{obs}})|_{\theta=\hat{\theta}}$ is the negative Hessian of the log-likelihood evaluated at the mode, often computed using automatic differentiation. While prediction intervals generated via the Laplace approximation can closely match those from more computationally intensive methods (see S&P2025), the method relies on the assumption that the posterior is locally Gaussian. This assumption may be violated for complex models, models with limited data, or simply for models with bounded parameters (e.g., $D \geq 0$; see Section 3.1) potentially leading to a poor representation of the true uncertainty.

However, all these likelihood-based approaches require an explicit specification of how the deterministic surrogate output relates to the noisy observational data—a critical modelling choice we examine next.

2.2.4 Noise Model Specification

The use of deterministic surrogates necessitates an explicit choice of how to model the relationship between the PDE output and the inherently noisy observations from the stochastic process. This choice profoundly impacts inference results, as the "noise" represents genuine stochastic variability in the agent-based system rather than measurement error.

S&P2025 systematically compare three noise models for relating observed counts to the PDE solution: Gaussian, Poisson, and binomial. The Gaussian noise model assumes observations are normally distributed around the model prediction with constant variance. While mathematically convenient, this leads to unphysical prediction intervals that include negative counts in low-density regions where the population density approaches zero, and fails to respect carrying capacity constraints in high-density regions. The Poisson noise model provides a more natural choice for count data, eliminating the need for an additional variance parameter and ensuring non-negative predictions. The binomial noise model is appropriate for exclusion processes where occupancy is bounded, naturally capturing saturation effects.

S&P2025's extensive comparisons reveal that noise model misspecification can significantly bias parameter estimates and lead to poorly calibrated prediction intervals. Quantitatively, the Gaussian model consistently produces unphysical negative predictions in low-density regions and encloses only 95% of data points, while the Poisson and binomial models better respect the discrete, non-negative nature of count data and achieve coverage rates of 99-100%. Similar conclusions regarding the superiority of binomial measurement models over standard additive Gaussian assumptions for biological count data have been demonstrated elsewhere [44].

2.3 Neural Posterior Estimation

2.3.1 Overview

NPE [35, 18, 9] takes a distinct approach to inference for simulator-based models within the broader simulation-based inference framework. Rather than relying on surrogate approximations or requiring explicit noise models, NPE learns the posterior distribution $p(\theta|Y)$ directly from simulations of the stochastic model. This is achieved through conditional density estimation using neural networks.

The key insight is that while the likelihood $p(Y|\theta)$ may be intractable for complex stochastic simulators, we can still readily generate samples from it by running the simulator. NPE leverages these (parameter, sample) pairs to train a neural network that approximates the inverse mapping—from an observation back to the distribution over parameters that could have produced it. This process is amortized: once the network is trained, it can perform posterior inference for any new observation in seconds rather than hours, without requiring additional simulations. For the class of stochastic biological models considered here, this approach directly addresses each limitation of classical methods: it circumvents intractable likelihoods, eliminates surrogate model bias, implicitly captures complex stochastic relationships without explicit noise models, and can process high-dimensional data through automatic feature extraction.

Figure 2 provides a schematic overview of the NPE workflow, illustrating the distinction between the computationally intensive training phase and the rapid inference phase. During training, parameters are sampled from the prior $p(\theta)$ and used to generate simulated data via the stochastic simulator $p(Y|\theta)$. These parameter-data pairs $\{(\theta_i, Y_i)\}_{i=1}^N$ are then used to train a neural density estimator $q_\phi(\theta|Y)$ that learns to approximate the posterior distribution. Once trained, the inference phase is nearly instantaneous: given new observed data Y_{obs}, the network directly evaluates the approximate posterior $p(\theta|Y_{\text{obs}})$ without requiring additional simulations.

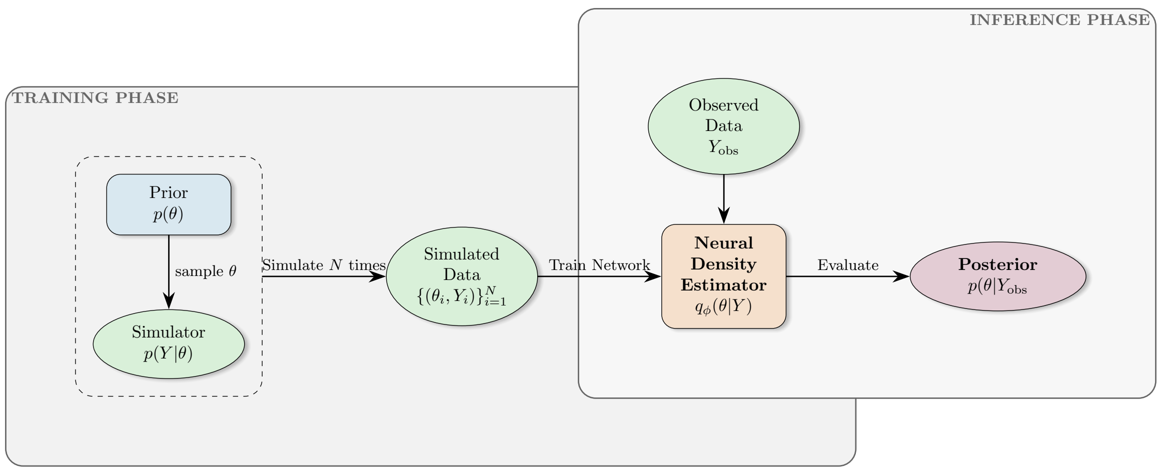


Figure 2: Schematic overview of the NPE workflow. The training phase (left) involves sampling parameters from the prior, generating simulated data, and training a neural density estimator on the resulting parameter-data pairs. The inference phase (right) uses the trained network to instantly evaluate the posterior for new observations without additional simulations.

2.3.2 Conditional Normalizing Flows

At the core of NPE are normalizing flows, a class of flexible neural architectures that learn complex probability distributions through invertible transformations. Normalizing flows have emerged as a powerful tool for density estimation in high-dimensional spaces [20, 36], offering significant advantages over traditional methods like mixture models or kernel density estimation when dealing with complex, multimodal distributions typical in biological inference problems. The underlying idea is to construct a complex target distribution by transforming samples from a simple base distribution (typically a standard Gaussian) through a series of invertible, differentiable mappings. This can be understood as a generalization of the change-of-variables technique extended to high dimensions through neural networks [26]. Normalizing flows leverage the representational power of deep learning while maintaining the mathematical rigour needed for proper density estimation.

Formally, let $u \in \mathbb{R}^d$ be a random variable with base density $p_u(u) = \mathcal{N}(0, I)$, where $I \in \mathbb{R}^{d\times d}$ denotes the identity matrix. The flow defines an invertible transformation $f_\phi : \mathbb{R}^d \to \mathbb{R}^d$ parametrized by neural network weights ϕ. For conditional density estimation, this transformation is conditioned on the observation: $\theta = f_\phi(u; x)$. The invertibility requirement ensures that we can both generate samples (forward pass) and evaluate densities (inverse pass). The key insight is that complex distributions can be learned by composing simple transformations, each of which maintains invertibility and efficient Jacobian computation. This allows the network to learn highly flexible posterior shapes while maintaining computational tractability, a crucial feature when inferring parameters in complex biological models where posteriors may exhibit non-Gaussian features such as multimodality, heavy tails, or curved degeneracies. The resulting approximate posterior density follows from the change of variables formula:

$$q_\phi(\theta|Y) = p_u(f_\phi^{-1}(\theta; Y)) \left|\det J_{f_\phi^{-1}}(\theta; Y)\right| \tag{4}$$

where $J_{f_\phi^{-1}}$ denotes the Jacobian of the inverse transformation. Intuitively, the determinant of the Jacobian corrects for the local change in volume caused by the transformation, ensuring the resulting density is properly normalized. To understand this geometrically: when the transformation f_ϕ stretches space in some region, the determinant accounts for this stretching to ensure probability mass is conserved [36]. Modern flow architectures (e.g., coupling flows [11], autoregressive flows [37]) are designed such that this determinant is computationally efficient to calculate—often reducing to a summation or product of scalar terms rather than requiring explicit $d \times d$ matrix determinant computation. For readers familiar with traditional MCMC-based inference in biological models, normalizing flows offer a complementary approach: rather than exploring the posterior through iterative sampling, they learn an explicit parametric approximation that enables fast posterior evaluation and sampling once trained [9]. This amortization—training once on simulated data, then instantly obtaining posteriors for new observations—makes them particularly attractive for scenarios requiring repeated inference, such as model selection, experimental design, or real-time data analysis.

2.3.3 Training Procedure

NPE training requires a dataset of parameter-data pairs generated from the simulator. The training procedure follows three main steps:

1. **Simulation:** Generate N training samples by drawing parameters from the prior $\theta_i \sim p(\theta)$ and simulating corresponding data $Y_i \sim p(Y|\theta_i)$ using the stochastic model (left section, Figure 2).
2. **Optimization:** Train the normalizing flow by maximizing the log-likelihood of parameters given their corresponding simulated data:
$$\mathcal{L}(\phi) = -\frac{1}{N}\sum_{i=1}^{N} \log q_\phi(\theta_i|Y_i)\,. \tag{5}$$
This objective is optimized using stochastic gradient descent (e.g., Adam [25]) with mini-batches of simulated pairs (θ_i, Y_i). During training, gradients are computed via back-propagation through the flow transformation $f_\phi(u; x)$, updating the network parameters ϕ until convergence. This learns the transformation that maps from a simple base distribution to the posterior distribution conditioned on the data (middle section, Figure 2).
3. **Inference:** For observed data Y_{obs}, obtain the approximate posterior $q_\phi(\theta|Y_{\text{obs}})$ by evaluating the trained network. Generate posterior samples by drawing $u \sim p_u(u)$ and applying the learned transformation $\theta = f_\phi(u; Y_{\text{obs}})$ (right section, Figure 2).

The training data requirements scale with problem complexity, but it is difficult in general to provide a quantitative recommendation on the number of required simulations to train the model. Typically at least $N = 10^4 - 10^6$ simulations suffice for moderate-dimensional parameter spaces ($d \leq 10$), though higher-dimensional problems may require larger datasets [10]. Crucially, this procedure is amortized; the computationally expensive training phase is performed only once, after which posterior inference for any new observation is nearly instantaneous. This represents a significant advantage over methods like ABC, which require fresh simulations for each new dataset. We provide quantitative comparisons in Section 3.4.

2.3.4 Sequential Neural Posterior Estimation

While the basic NPE algorithm is powerful, it can be inefficient when the prior is much broader than the posterior. Sequential Neural Posterior Estimation (SNPE) [35, 18] addresses this through an iterative refinement process that concentrates computational effort on parameter regions consistent with the observed data.

The sequential algorithm proceeds as follows:

1. **Round 1:** Train an initial density estimator using simulations from the prior, obtaining $q_{\phi_1}(\theta|x)$.
2. **Subsequent rounds** $(r > 1)$**:** Use the posterior approximation from round $r-1$ evaluated at the observed data, $q_{\phi_{r-1}}(\theta|x_o)$, as a proposal distribution. Generate new simulations $\theta \sim q_{\phi_{r-1}}(\theta|x_o)$, $x \sim p(x|\theta)$.
3. **Importance weighting correction:** Train a new density estimator $q_{\phi_r}(\theta|x)$ on the focused simulations, with importance weighting to correct for the proposal distribution bias:
$$\mathcal{L}_r(\phi) = -\frac{1}{N_r}\sum_{i=1}^{N_r} w_i \log q_\phi(\theta_i|x_i) \tag{6}$$
where $w_i = p(\theta_i)/q_{\phi_{r-1}}(\theta_i|x_o)$ corrects for sampling from the proposal rather than the prior.

SNPE is most beneficial when the prior is substantially broader than the posterior, which is common in biological applications where parameter ranges are often set conservatively. The method typically converges within 2–4 rounds, providing substantial efficiency gains over standard NPE while maintaining accuracy. In this work we exclusively use the sequential version of NPE.

2.3.5 Implementation Details

All NPE analyses in this work use the `sbi` Python package [51], which provides a well-documented implementation of state-of-the-art simulation-based inference algorithms. We employ neural spline flows [12] as our density estimator architecture, which offer excellent expressivity while maintaining computational efficiency. Additional implementation details can be found in the project GitHub repository [2].

We use broad uniform priors covering the full parameter space: $P \sim \mathcal{U}(0.0, 1.0)$ and $U \sim \mathcal{U}(0.01, 1.0)$. These priors are determined by the physical constraints of the model: U and P are both probabilities and so must lie in $[0, 1]$. The lower bound of 0.01 for U excludes the degenerate case of zero initial agents while remaining sufficiently permissive. In the absence of strong prior information about parameter values, these uniform priors represent a weakly informative choice that allows the data to dominate the posterior inference. For NPE, the prior serves the dual role of defining the training distribution from which simulation parameters are drawn, and entering into the posterior distribution via Bayes' theorem [15, 16].

2.4 Data Representations and CNN Architecture

A key feature of the barrier assay experimental design is the reduction of two-dimensional spatial data to one-dimensional measurements. Since the initial macroscopic density is uniform in the vertical (y) direction and boundary conditions preserve this symmetry, the resulting agent distribution remains independent of vertical location on average. This motivates quantifying the system by counting the number of agents per column:

$$N_i(t) = \sum_{j=1}^{J} n_{i,j}(t), \tag{7}$$

where $n_{i,j}(t)$ denotes the integer number of agents at lattice site (i, j) at time t, and J is the number of lattice sites in the vertical direction. This dimensional reduction from 2D spatial data to 1D column counts is standard in cell biology scratch assays [21], as it simplifies the characterization of spatio-temporal spreading while retaining the stochastic fluctuations inherent to the discrete process, as evident in Figure 1(c)–(d). The continuous analogue of the column count N_i under the PDE model is $Hu(x_i, t)$, where x_i denotes the horizontal position of the i-th column and H is the lattice height. Figure 1(c)–(d) compares the stochastic count data (blue dots) with this PDE solution (red curves), revealing that while the PDE captures mean trends, substantial stochastic fluctuations remain.

For inference on 2D spatial data, we treat each lattice configuration from the simulator as a 2D histogram (or count-based image), where each pixel's value represents the integer number of agents $n_{i,j}(t)$ at the corresponding lattice site. To learn informative features directly from this high-dimensional input, we replace the simple feed-forward network with a CNN [28, 27, 17]. The CNN acts as a learnable feature extractor embedded within the NPE pipeline. It uses a series of convolutional and pooling layers to progressively downsample the spatial data into a low-dimensional embedding. This process captures salient spatial features, such as clustering patterns and local agent densities, while preserving translational invariance. The learned embedding is then passed to the remainder of the network to estimate the posterior distribution. This end-to-end training ensures that the features extracted by the CNN are maximally informative for parameter inference.

Table 2 summarises the key hyperparameters for both the 1D and 2D NPE configurations. A detailed specification of the CNN architecture and full implementation details can be found in the project GitHub repository.

3 Results

We aim to infer two key parameters that govern the model's behaviour: the initial occupation probability, U, and the effective diffusion coefficient, D. For the synthetic data considered here, we use ground truth values $U = 0.3$ and $D = 0.175$.

[2] https://github.com/tomkimpson/NPE_for_RW_inference

Hyperparameter	1D (Column Counts)	2D (Spatial / CNN)
Input dimension	200	$50 \times 200 = 10{,}000$
Density estimator	Neural Spline Flows	Neural Spline Flows
Coupling layers	8	5
Hidden features	128	256
Embedding network	Feed-forward	CNN (4 conv. stages)
Learning rate	10^{-4}	5×10^{-5}
Batch size	128	512
Max epochs	300	150
Early stopping patience	10 epochs	40 epochs
SNPE rounds	10	8
Simulations per round	2,000	6,000
Convergence threshold	0.001	0.01

Table 2: Summary of key hyperparameters for the NPE pipeline in the 1D and 2D configurations. All models were trained on an NVIDIA A100-SXM4-80GB GPU using the `sbi` library with a PyTorch backend.

3.1 Classical Inference Baseline

Figure 3a shows the posterior obtained via ABC. This method operates directly on the stochastic model but relies on one-dimensional summary statistics (column counts), leading to a notably broad posterior. This diffuseness reflects both the inherent stochasticity of the system and the information loss from summarizing the data, showcasing the difficulty in precisely constraining parameters even with a high computational budget. In contrast, Figure 3b displays the Gaussian approximation to the posterior (Laplace approximation) derived from maximizing the likelihood of a deterministic surrogate. This approach is computationally efficient and yields a much tighter posterior. However, this apparent precision is entirely conditional on the fidelity of the surrogate; any mismatch between the surrogate and the true stochastic process introduces a systematic bias not reflected in the narrow posterior. Finally, Figure 3c shows the posterior sampled using MCMC with the surrogate model and an assumed Gaussian noise model. This provides a more flexible characterization of uncertainty. Crucially, this method requires inferring an additional nuisance parameter, σ, which quantifies the noise variance. Figure 3c shows the joint posterior over the model parameters and this additional noise term. While comprehensive, this result is still fundamentally limited by the choice of surrogate and the specified noise model.

Collectively, these results highlight the core dilemma: direct methods like ABC are computationally demanding and often imprecise, while surrogate-based methods are faster but introduce hard-to-quantify systematic biases that cannot be reduced by collecting more data.

3.2 NPE with 1D Summary Statistics

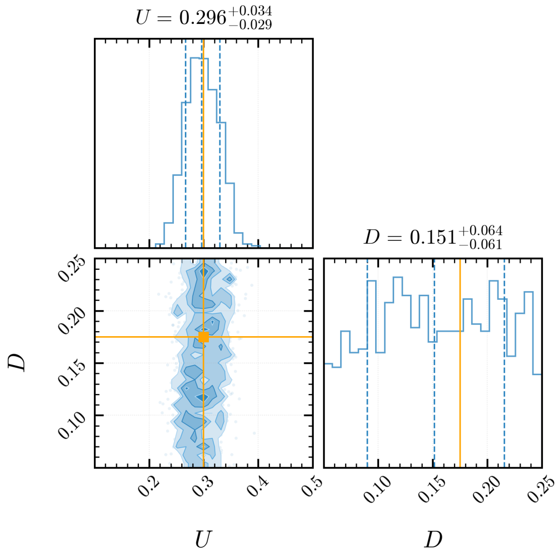
We demonstrate NPE's effectiveness by applying it to the same synthetic barrier assay data used in Section 3.1. To ensure direct comparison with the classical methods, we first implement NPE using identical 1D column count summary statistics N_i, representing the number of agents in each vertical column of the lattice.

Following the protocol of S&P2025, we generate synthetic observations from a stochastic random walk model with true parameters $P = 0.7$ and $U = 0.3$, corresponding to diffusivity $D \approx P/4 = 0.175$. The resulting data consists of a 201-dimensional vector of column counts, corresponding to the 201 discrete lattice columns from $x = -100$ to $x = +100$. We perform inference directly on the stochastic model parameters P and U, then transform to D via the relationship $D = P/4$ for comparison with PDE-based approaches.

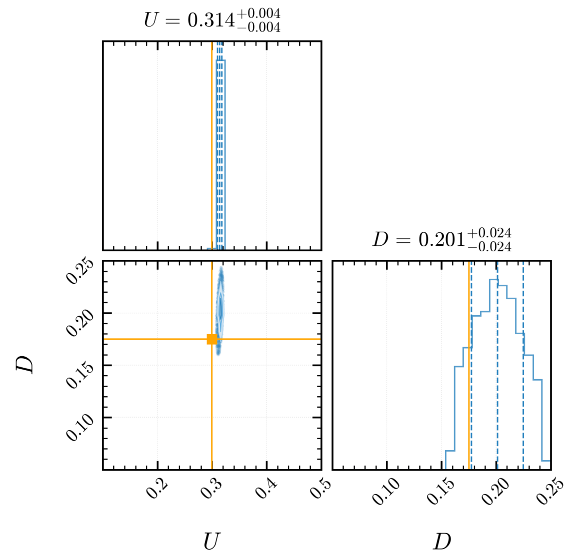
3.2.1 Parameter Inference Results

Figure 4 presents the posterior distribution obtained via NPE. NPE recovers a posterior distribution with median $U = 0.298^{+0.009}_{-0.008}$ and $D = 0.164^{+0.024}_{-0.024}$. The results demonstrate excellent agreement with the true parameter values ($U = 0.3$, $D = 0.175$).

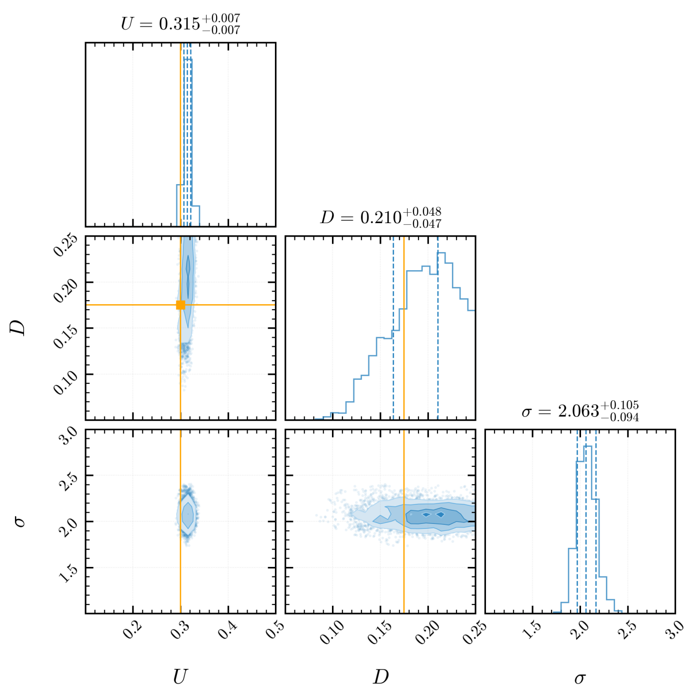
This approach yields several key advantages over classical methods. The NPE posterior shows substantially improved precision, than a posterior obtained from a comparable ABC analysis, especially for the diffusion parameter D, all



(a) Approximate posterior from ABC applied directly to the stochastic model. Computational constraints and information loss lead to a diffuse posterior for D.



(b) Gaussian approximation of the posterior (Laplace approximation) via MLE on a deterministic surrogate model. This approach is computationally efficient but conditional on the surrogate's fidelity.



(c) Posterior distribution from MCMC using the surrogate model with an assumed Gaussian noise model. This method jointly infers the model parameters alongside an additional noise variance parameter, σ.

Figure 3: Comparison of posterior distributions obtained using classical inference methods for a scratch assay model. The diagonal panels show the marginalized one-dimensional posteriors for each parameter, while the off-diagonal panel shows the joint two-dimensional posterior illustrating their covariance. The subtitles above each one-dimensional subplot indicate the posterior median and the central 68% credible interval. The vertical dashed lines mark 16th, 50th, and 84th percentiles of each one-dimensional posterior, corresponding to the median (central line) and the bounds of the central 68% credible interval. The orange lines indicate the true parameter values. Each approach demonstrates distinct trade-offs between computational efficiency, precision, and model assumptions.

without the need for manual tolerance tuning. Furthermore, because NPE learns directly from the stochastic simulator, it avoids the systematic bias of mean-field approximations and produces a posterior that is not artificially narrow. The resulting credible intervals accurately reflect the natural stochastic variance of the data, a feature often lost when using deterministic surrogate models, cf. Fig 3. Once the initial network training is complete, posterior samples for any new observation can be obtained almost instantly.

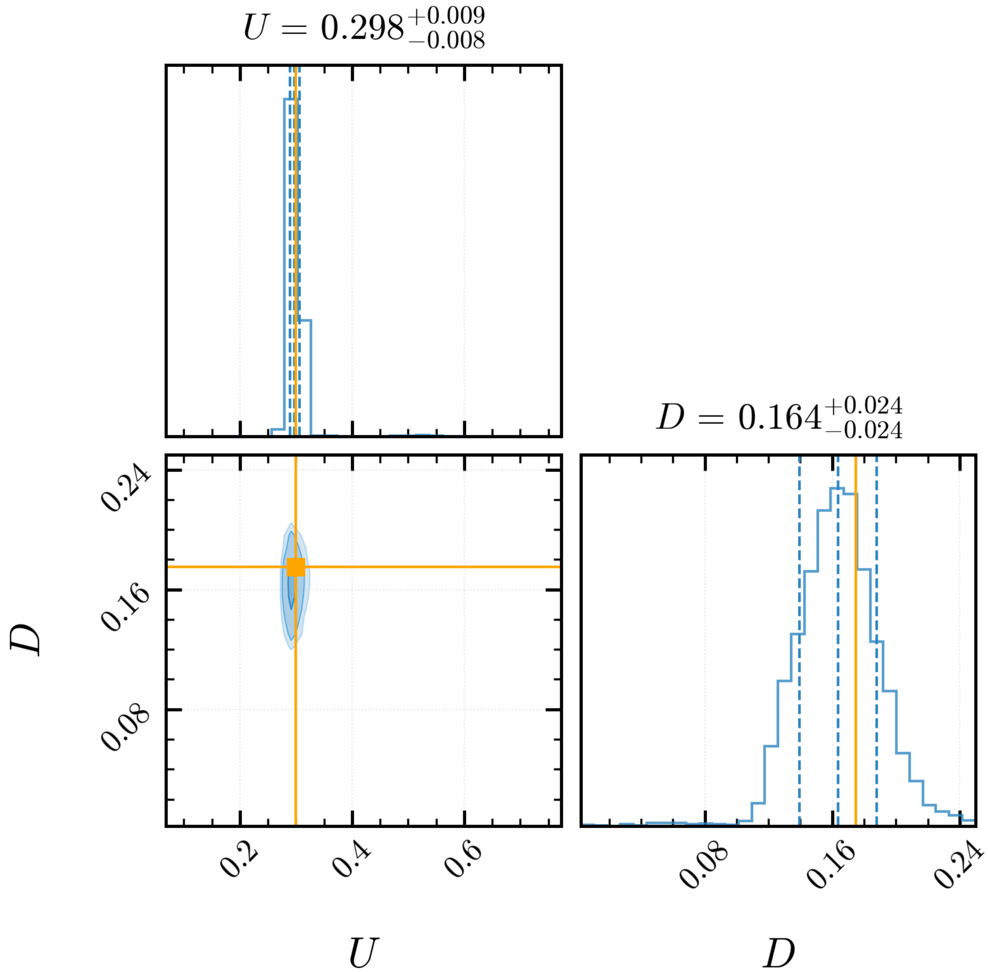


Figure 4: Posterior distribution from NPE using 1D column count data. The figure is arranged as described in Figure 3. The true parameter values (orange cross) are contained within the 68% credible interval

3.2.2 Predictive Performance

Beyond parameter estimation, we evaluate NPE's predictive capabilities through posterior predictive sampling. For each posterior sample $\theta^{(j)}$, we generate predicted data $\tilde{x}^{(j)} \sim p(x|\theta^{(j)})$ using the stochastic simulator. The collection of predictions forms empirical prediction intervals that naturally account for both parameter uncertainty and model stochasticity.

As shown in Figure 5, the resulting NPE-based predictions are well-calibrated, with 94.8% of true observations falling within the 95% prediction intervals. A key advantage of this simulator-native approach is its physical consistency; the prediction intervals respect the non-negative, discrete nature of the count data without requiring an explicit, and potentially erroneous, noise model. Furthermore, the framework naturally captures heteroscedastic uncertainty, with the prediction intervals appropriately widening in regions of high agent density where stochastic variability is greatest.

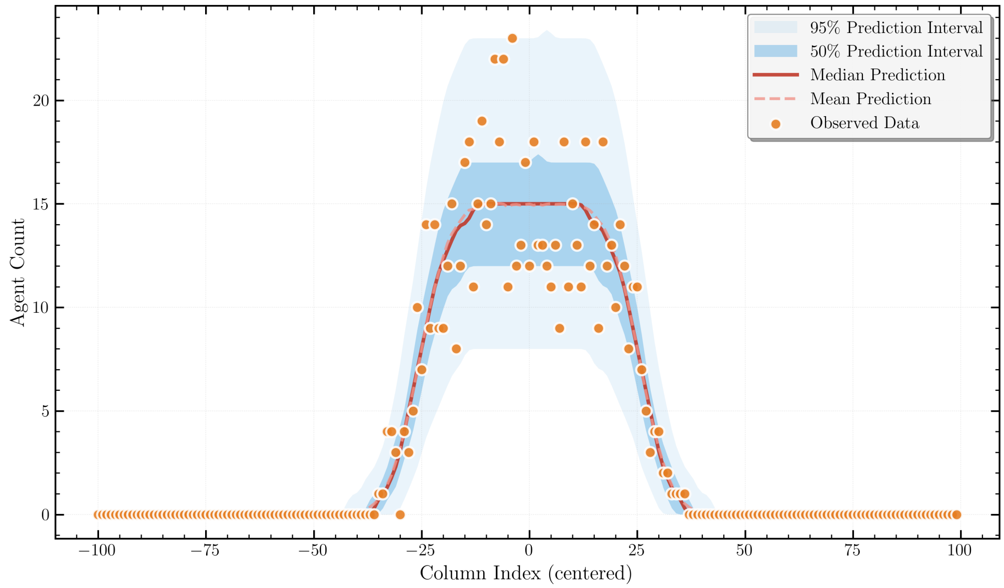


Figure 5: Posterior predictive checks from NPE using 1D column count data. The observed data (orange circles) is overlaid on predictions generated by simulating the model with parameters drawn from the inferred posterior distribution. The median (solid red line) and mean (dashed red line) of the predictions are shown, along with the 50% (dark blue) and 95% (light blue) prediction intervals. For visualization purposes, a Gaussian kernel has been applied to smooth the predictive distributions. An appropriate proportion of the observed data points lie within the 95% prediction interval, indicating that the calibrated model is consistent with the data.

3.3 NPE with 2D Spatial Data

While column counts provide a natural summary of the spreading process, they necessarily discard information about the vertical distribution of agents. In principle, the spatial correlation structure within the 2D configuration contains additional information about the underlying parameters. For instance, clustering patterns, local density fluctuations, and spatial correlations may help distinguish between different combinations of P and U that produce similar column counts, cf. parameter identifiability [3]. Traditional inference methods require hand-crafted summary statistics, making it challenging to leverage such high-dimensional spatial information. NPE, however, can learn directly from raw data through automatic feature extraction. While the simple random walk models considered here are likely well-captured by 1D summary statistics, this approach demonstrates the broader potential of NPE for more complex biological systems where spatial structure is critical. For example, models incorporating agent-agent interactions, complex boundary conditions, spatially heterogeneous environments, or multi-species dynamics would generate spatial patterns that cannot be adequately summarized by simple column counts. In such cases, the ability to learn directly from full spatial data becomes essential for accurate parameter inference.

3.3.1 Experimental Setup

The experimental setup remains identical to the 1D case, using the same synthetic observation generated with true parameters $P = 0.7$ ($D = 0.7/4$) and $U = 0.3$. However, instead of using the 201-dimensional vector of column counts as input, the network now receives the full 50×200 binary image representing the agent locations on the lattice. We use the same broad uniform priors for P and U as described in Section 2.3.

3.3.2 Parameter Inference Results

Figure 6 shows the joint posterior distribution inferred from the 2D spatial data. The CNN-based approach recovers a posterior with means $D = 0.169^{+0.026}_{-0.023}$ and $U = 0.290^{+0.008}_{-0.09}$. As summarised in Table 3, leveraging the full spatial

[3] Although parameter identifiability is not a significant concern for the simple model we consider, additional spatial features are of interest for disentangling parameters in more complex models that incorporate processes like cell-cell adhesion or proliferation. We note that for longer times even this simple model becomes non-identifiable in the diffusion coefficient D.

data yields a posterior with a precision that is highly comparable to that obtained using curated 1D summary statistics, cf. Section 3.2.

This result demonstrates that the CNN can automatically learn features from the raw data that are as informative as the carefully chosen summary statistics. This highlights a critical trade-off: the increased computational cost of training the CNN (Table 3) must be weighed against the major advantage of avoiding manual feature engineering. For more complex models where optimal summary statistics are unknown, this direct-from-pixels approach may be advantageous for accurate inference.

3.3.3 Predictive Performance

We assess the predictive performance of the 2D-trained model using the same posterior predictive checking procedure as in Section 3.2.2. As shown in Figures 7 and 8, the model generates predictions that are highly consistent with the observed data. The calibrated model accurately captures the spatial characteristics of the agent-based simulation, producing prediction intervals that are well-calibrated and physically consistent with the discrete nature of the data. Importantly, the framework captures the input-dependent (heteroscedastic) uncertainty, something that standard MLE under an i.i.d. noise assumption (used in much of the mathematical biology literature) cannot represent, often leading to miscalibrated, homoscedastic intervals even when the underlying variability is spatially varying. As in the 1D case, the method remains robust when applied to high-dimensional (2D) spatial inputs.

In the 2D (CNN) results we also note a modest separation between the posterior predictive mean and median, consistent with a mild right-skew in the posterior for D. We attribute the observed skew primarily to structural features of the inference problem—namely a slight U–D trade-off, support constraints ($D \geq 0$ with $D = P/4$), and the non-linear parameter-to-observable map—together with the fact that the CNN accesses richer spatial cues (e.g., front roughness and local clustering) that are averaged out in 1D summaries and therefore make such asymmetries more apparent. We cannot entirely rule out minor artefacts arising from the higher-dimensional learning task.

Data Type	**Posterior for U**	**Posterior for D**
1D Column counts	$0.298^{+0.009}_{-0.008}$	$0.164^{+0.024}_{-0.024}$
2D Spatial data	$0.290^{+0.008}_{-0.009}$	$0.169^{+0.026}_{-0.023}$

Table 3: Comparison of NPE performance using different data representations. Values show posterior medians with 68% credible intervals (x^{+a}_{-b}). Inference using the 2D spatial data yields comparable posterior precision to the 1D summary statistics.

3.4 Computational Performance

To provide a practical comparison of computational costs, Table 4 summarises representative runtimes for the different inference methods. These figures are illustrative of a typical workflow and are not the result of systematic hyper-parameter optimisation. For example, the NPE training time scales roughly linearly with the number of sequential rounds: the reported 1D case used ten rounds, so halving or doubling this number would proportionally reduce or increase the runtime, with an uninvestigated trade-off in posterior accuracy. Similarly, ABC runtime depends directly on the number of simulations generated (here, 10^5). Other hyper-parameters (e.g. number of network epochs, batch size, learning rate, width/depth of the embedding network) also have a direct influence on both runtime and accuracy, but were fixed to reasonable defaults rather than tuned for performance. We also note the difference in hardware: the computationally expensive NPE training phase was performed on a GPU, whereas the other methods ran on a standard CPU.

Two distinct computational strategies emerge from this comparison. Surrogate-based approaches are extremely fast because they entirely bypass repeated stochastic simulations, relying instead on precomputed or deterministic approximations of the model. In contrast, the NPE framework demands a substantial one-time computational investment during training but then performs posterior inference for new observations almost instantaneously. This amortised structure makes NPE particularly attractive for scenarios requiring repeated or real-time inference across many datasets.

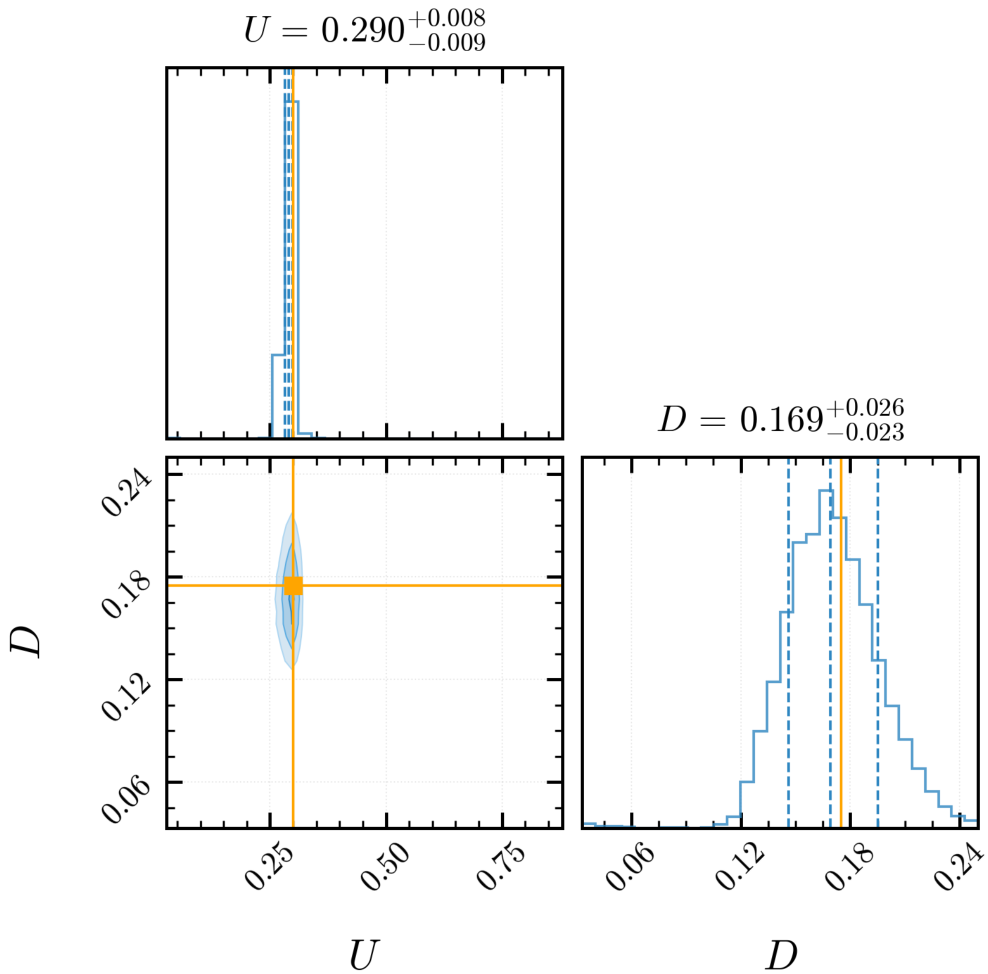


Figure 6: As Figure 4, with posteriors generated from the NPE with CNN embedding networks, using full 2D spatial data.

Method	**Model / Data Type**	**Training Time (s)**	**Inference Time (s)**
ABC	Stochastic Simulator (1D Data)	–	∼1200
ABC	Surrogate Model (1D Data)	–	∼2
MCMC	Surrogate Model (1D Data)	–	∼1
NPE	**Stochastic Simulator (1D Data)**	**∼6400**	$<$ **0.1**
NPE + CNN	**Stochastic Simulator (2D Data)**	**∼9800**	$<$ **0.1**

Table 4: Representative computational runtimes for various inference methods. NPE requires a significant one-time training cost, after which inference is near-instantaneous. The NPE training was conducted on a GPU, while other methods were run on a CPU.

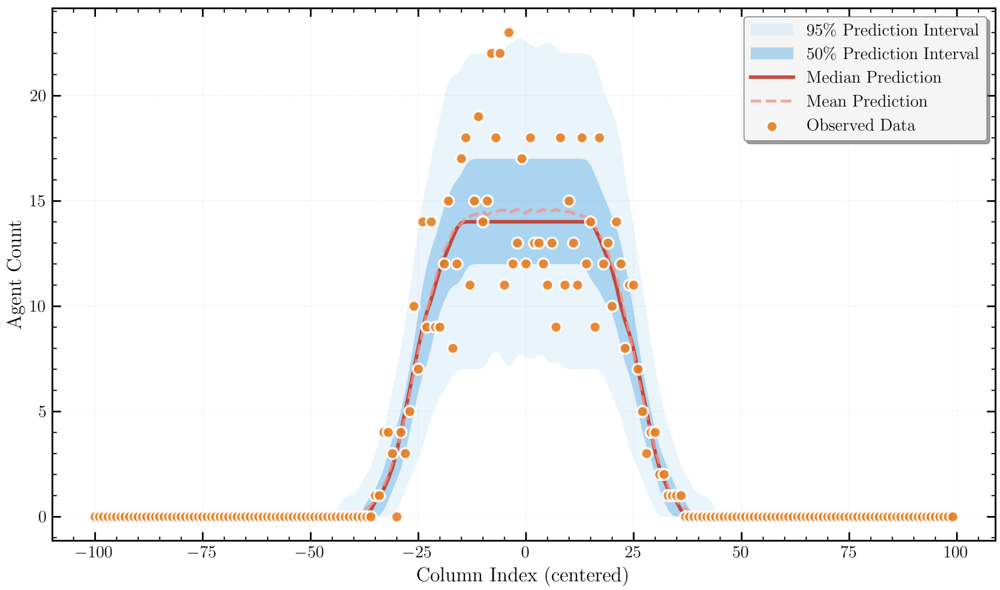


Figure 7: Posterior predictive check for the 2D spatial data model. The format is identical to Figure 5. The predictions are well-calibrated and consistent with the observed data.

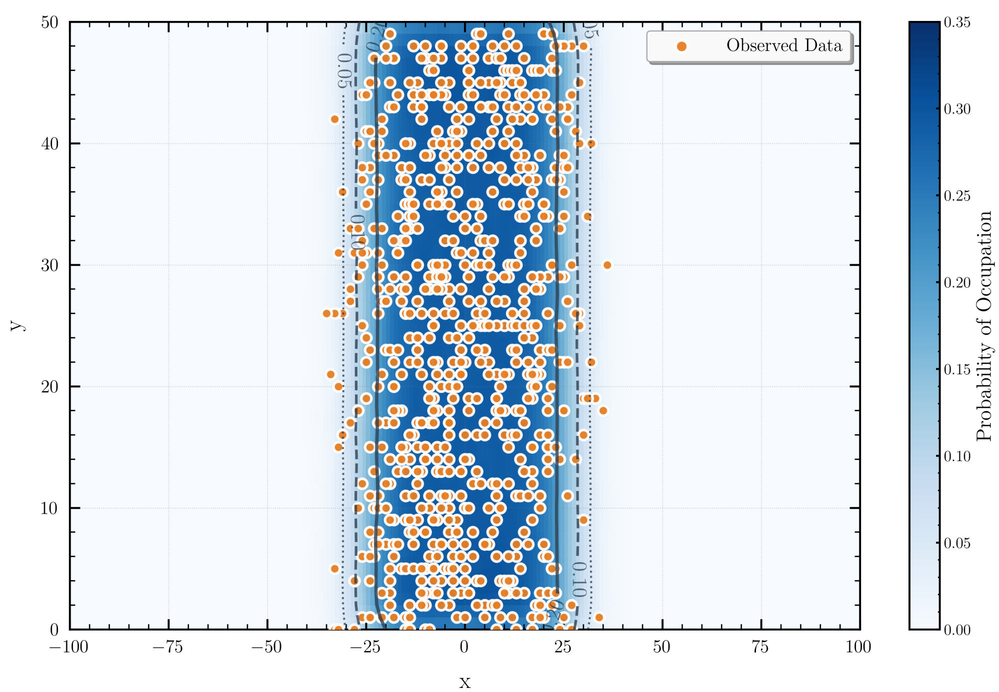


Figure 8: Posterior predictive check for the 2D spatial data model, showing the full observed data points (orange) overlaid on the predicted occupancy probability from the NPE model (blue heat-map). The grey contours delineate iso-probability curves at probability levels of 0.05, 0.10, and 0.20 (dotted, dashed, solid, respectively).

4 Discussion

4.1 Advantages and Limitations of Neural Posterior Estimation

NPE offers a robust and flexible framework for parameter inference, directly addressing several challenges that have limited the practical application of stochastic agent-based models. NPE eliminates the need for surrogate models by learning directly from the full stochastic simulator. This direct approach circumvents the systematic bias introduced by mean-field approximations and removes the challenging requirement of specifying explicit noise models that connect deterministic surrogates to stochastic observations. The method's ability to process raw high-dimensional data through automatic feature extraction represents another advantage, avoiding the information loss inherent in manually crafted summary statistics while enabling the analysis of complex spatial patterns that would be difficult to characterize a priori. The amortized nature of NPE provides substantial computational advantages once training is complete. After the initial investment in generating training data and optimizing the neural network, posterior inference for new observations requires only milliseconds of network evaluation rather than hours (or more) of simulation-based methods like ABC. This efficiency enables real-time analysis and interactive exploration of the parameter space that would be prohibitive with classical approaches. Furthermore, normalizing flows can capture complex, multimodal posterior geometries that arise naturally from model non-identifiabilities without requiring restrictive parametric assumptions about posterior shape.

However, NPE also introduces new challenges and limitations that practitioners must carefully consider. The method's appetite for training data can be substantial, particularly for high-dimensional parameter spaces where the curse of dimensionality becomes prohibitive. Problems with more than 15–20 parameters may require simulation budgets that exceed practical computational limits. Additionally, NPE performance can degrade significantly when test observations fall outside the training distribution, making careful prior specification and robust training data generation critical for reliable inference.

The neural network architecture introduces another layer of complexity, as design choices regarding depth, width, and flow type can dramatically impact performance. Unlike classical methods with well-established theoretical foundations, NPE provides limited formal guarantees about convergence or approximation quality. This black-box nature can make diagnosis of poor performance challenging and requires practitioners to develop intuition about network behaviour and training dynamics.

NPE is one of several neural approaches to simulation-based inference [9, 10]. Neural Ratio Estimation (NRE) learns the likelihood-to-evidence ratio and requires MCMC sampling to obtain posterior samples [19], while Neural Likelihood Estimation (NLE) learns a surrogate likelihood function that must likewise be combined with MCMC [38]. We chose NPE because it directly produces the posterior density, enabling amortized inference without any additional sampling step. Once trained, each new observation yields posterior samples via a single forward pass, which is particularly advantageous in our barrier assay setting where many experimental replicates must be analysed. Benchmark studies in low-to-moderate-dimensional problems show that NPE performs competitively with NRE and NLE in terms of posterior accuracy [30], while its amortized design avoids the per-observation MCMC cost incurred by the ratio- and likelihood-based alternatives. We note that NRE or NLE may be preferable for problems where the likelihood-to-evidence ratio is simpler to learn than the full posterior, or where sequential refinement around a single observation is prioritised over amortization across many observations.

4.2 Role of Spatial Information

Our results highlight a critical trade-off between methodological complexity and computational cost. For the simple random walk model, well-chosen summary statistics provide comparable precision to the full spatial analysis at a fraction of the cost, making them a highly effective and efficient choice. However, this should not obscure the broader potential of spatial inference approaches; the cost-benefit calculation shifts decisively toward spatial approaches for more complex biological systems where crucial information is encoded in patterns that resist simple summarization, or where informative statistics are not known a priori.

In many realistic scenarios, crucial mechanistic information is encoded in intricate spatial patterns that are lost in simple one-dimensional projections. For example, models incorporating agent-agent interactions through exclusion effects, chemotaxis, or cell-cell adhesion generate complex formations that are lost when compressed into simple one-dimensional projections. Similarly, heterogeneous environments with spatially varying growth rates, physical barriers, or chemical gradients create complex spatial signatures that contain crucial parameter information. Multi-species dynamics involving competition, predation, or cooperation also produce emergent spatial organization that encodes information about underlying mechanisms. Collective behaviours such as swarming, flocking, or tissue morphogenesis involve coordinated movements that are fundamentally spatial phenomena requiring spatial analysis for proper characterization. In these more complex scenarios, CNN approaches would preserve information that would otherwise be discarded by traditional summary statistics.

A critical aspect of applying deep learning models like CNNs in a scientific context is interpretability. While these networks are often treated as "black boxes", it is both possible and important to investigate the features they learn to solve a given inference problem. In principle, techniques such as activation mapping [33] or saliency maps [34] could be used to visualize the spatial features our CNN learns to associate with different parameter values. One might expect the network to develop sensitivity to biologically relevant patterns like the sharpness of the spreading front, local cell density gradients, or the degree of cell clustering. By identifying these learned features, a researcher could validate that the network's reasoning aligns with established biological theory, or discover novel spatial statistics that are informative but not immediately obvious to a human observer. While a detailed feature analysis is beyond the scope of this work, we highlight its importance for future applications. The ability to not only perform inference but also to extract the learned representations that enable it is a key advantage of this approach.

The ability to perform inference with stochastic models on spatiotemporal data has direct implications for experimental design. Traditional protocols often reduce rich data from sources like time-lapse microscopy to simplified measurements. This is a compromise driven by past computational limits, not biological necessity. Our framework removes this barrier, enabling direct analysis that preserves the full spatial information. This should encourage the strategic collection and retention of rich datasets, which is crucial for improving parameter identifiability in complex systems where spatial structure encodes mechanistic information.

4.3 Computational Scalability and Future Directions

The primary cost for any NPE approach is the upfront simulation budget, which can become substantial for models with high-dimensional parameter spaces ($>$ 15–20 parameters). On top of this, the CNN adds specific challenges: (i) memory requirements scale quadratically with spatial resolution, potentially limiting application to very high-resolution imaging data without careful optimization; (ii) architecture selection for CNN embedding networks requires problem-specific tuning of depth, width, and kernel sizes, adding complexity to the analysis pipeline; (iii) the combined optimization of feature extraction and density estimation can be more challenging than simpler approaches, requiring careful hyper-parameter selection and potentially longer training times.

Despite these challenges, several technological trends suggest that spatial inference approaches will become increasingly practical. Advances in GPU computing and specialized hardware for neural network training continue to reduce computational barriers. Improved network architectures and training algorithms may reduce the simulation budgets required for reliable inference. Transfer learning approaches could enable pre-trained spatial feature extractors to be adapted to new biological problems with reduced computational cost.

The framework developed here provides a foundation that can evolve with these technological advances while addressing increasingly complex biological questions. Future extensions might incorporate three-dimensional spatial data, temporal dynamics, or multi-modal observations combining spatial patterns with biochemical measurements. As biological modelling advances toward more realistic representations of complex systems, the ability to learn directly from high-dimensional, spatially structured data will become essential for extracting quantitative insights from biological observations.

Parameter inference for spatially explicit stochastic biological models has long been hampered by a fundamental tension: the need for computational tractability versus the desire for biological realism. Classical approaches have required researchers to choose between computationally demanding likelihood-free methods that operate directly on stochastic simulators, or efficient likelihood-based techniques that rely on potentially unfaithful surrogate models and explicit noise specifications. This work demonstrates that NPE provides a principled path beyond this dichotomy.

We have shown that NPE, augmented with modern deep learning architectures, addresses each of the canonical challenges that have historically limited stochastic biological modelling. By learning posteriors directly from simulator output, the framework circumvents intractable likelihoods while avoiding the systematic biases introduced by mean-field approximations. The method's implicit learning of complex stochastic relationships eliminates the need for explicit noise model specification. Once trained, the amortized nature of NPE provides near-instantaneous posterior estimates, overcoming the prohibitive per-inference costs of traditional likelihood-free methods. By integrating CNNs as learned feature extractors, the framework can process high-dimensional spatial data directly, removing the need for hand-crafted summary statistics that risk substantial information loss. As computational resources continue to expand and neural network architectures become more sophisticated, we anticipate that simulation-based inference frameworks like NPE will become a standard approach for parameter estimation in stochastic biological systems. We provide an open-source implementation of our complete pipeline[4] to facilitate adoption and enable the community to extend these methods to increasingly complex spatially-structured models.

[4] github.com/tomkimpson/NPE_for_RW_inference

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